CSC 220.03

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Assignment 04

Due Date & Time: 04-06-2022 at 11:55 PM

Assignment 04

Part A:

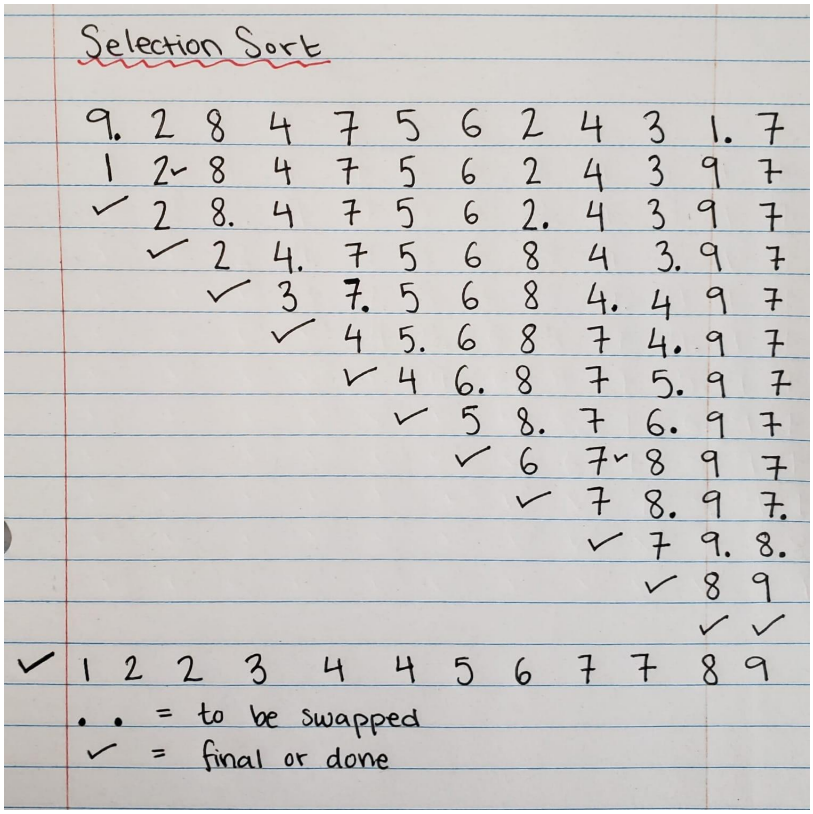
1. **Selection Sort**

```
SelectionSortDriver
/Library/Java/JavaVirtualMachines/
List Before Selection Sort:
9 2 8 4 7 5 6 2 4 3 1 7
Is sorted: false

1 2 8 4 7 5 6 2 4 3 9 7
1 2 8 4 7 5 6 2 4 3 9 7
1 2 2 4 7 5 6 8 4 3 9 7
1 2 2 3 7 5 6 8 4 4 9 7
1 2 2 3 4 5 6 8 7 4 9 7
1 2 2 3 4 4 6 8 7 5 9 7
1 2 2 3 4 4 5 8 7 6 9 7
1 2 2 3 4 4 5 6 7 8 9 7
1 2 2 3 4 4 5 6 7 8 9 7
1 2 2 3 4 4 5 6 7 7 9 8
1 2 2 3 4 4 5 6 7 7 8 9

List After Selection Sort:
1 2 2 3 4 4 5 6 7 7 8 9
Is sorted: true


Process finished with exit code 0
```



Selection Sort

9.	2	8	4	7	5	6	2	4	3	1.	7
1	2✓	8	4	7	5	6	2	4	3	9	7
✓	2	8.	4	7	5	6	2.	4	3	9	7
	✓	2	4.	7	5	6	8	4	3.	9	7
		✓	3	7.	5	6	8	4.	4	9	7
			✓	4	5.	6	8	7	4.	9	7
				✓	4	6.	8	7	5.	9	7
					✓	5	8.	7	6.	9	7
						✓	6	7✓	8	9	7
							✓	7	8.	9	7.
								✓	7	9.	8.
									✓	8	9
										✓	✓
✓ 1	2	2	3	4	4	5	6	7	7	8	9

. . = to be swapped

✓ = final or done

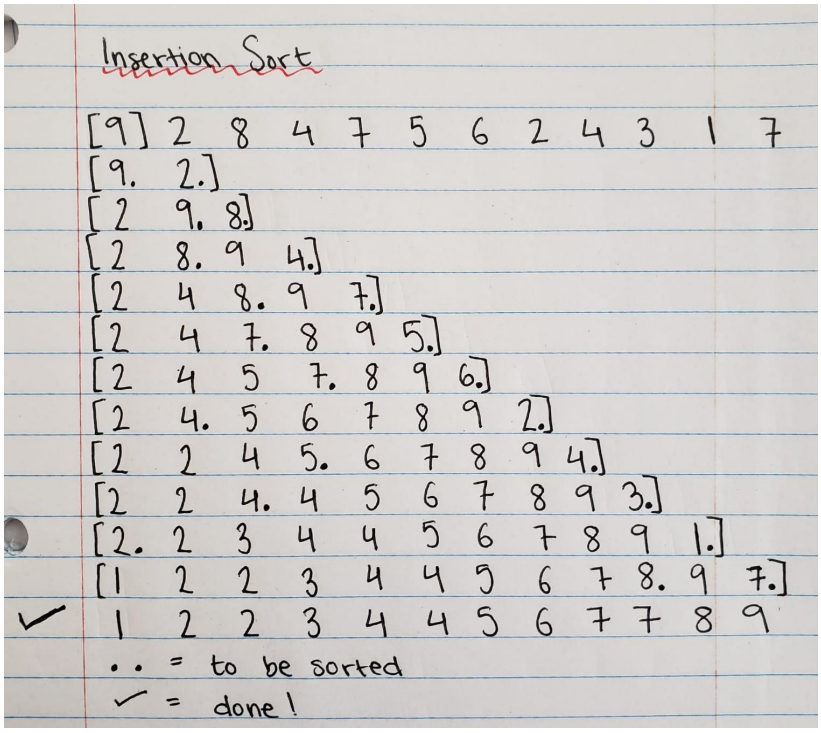
2. Insertion Sort

```
InsertionSortDriver
/Library/Java/JavaVirtualMachines/
List Before Insertion Sort:
9 2 8 4 7 5 6 2 4 3 1 7
Is sorted: false

2 9 8 4 7 5 6 2 4 3 1 7
2 8 9 4 7 5 6 2 4 3 1 7
2 4 8 9 7 5 6 2 4 3 1 7
2 4 7 8 9 5 6 2 4 3 1 7
2 4 5 7 8 9 6 2 4 3 1 7
2 4 5 6 7 8 9 2 4 3 1 7
2 2 4 5 6 7 8 9 4 3 1 7
2 2 4 4 5 6 7 8 9 3 1 7
2 2 3 4 4 5 6 7 8 9 1 7
1 2 2 3 4 4 5 6 7 8 9 7
1 2 2 3 4 4 5 6 7 7 8 9

List After Insertion Sort:
1 2 2 3 4 4 5 6 7 7 8 9
Is sorted: true


Process finished with exit code 0
```



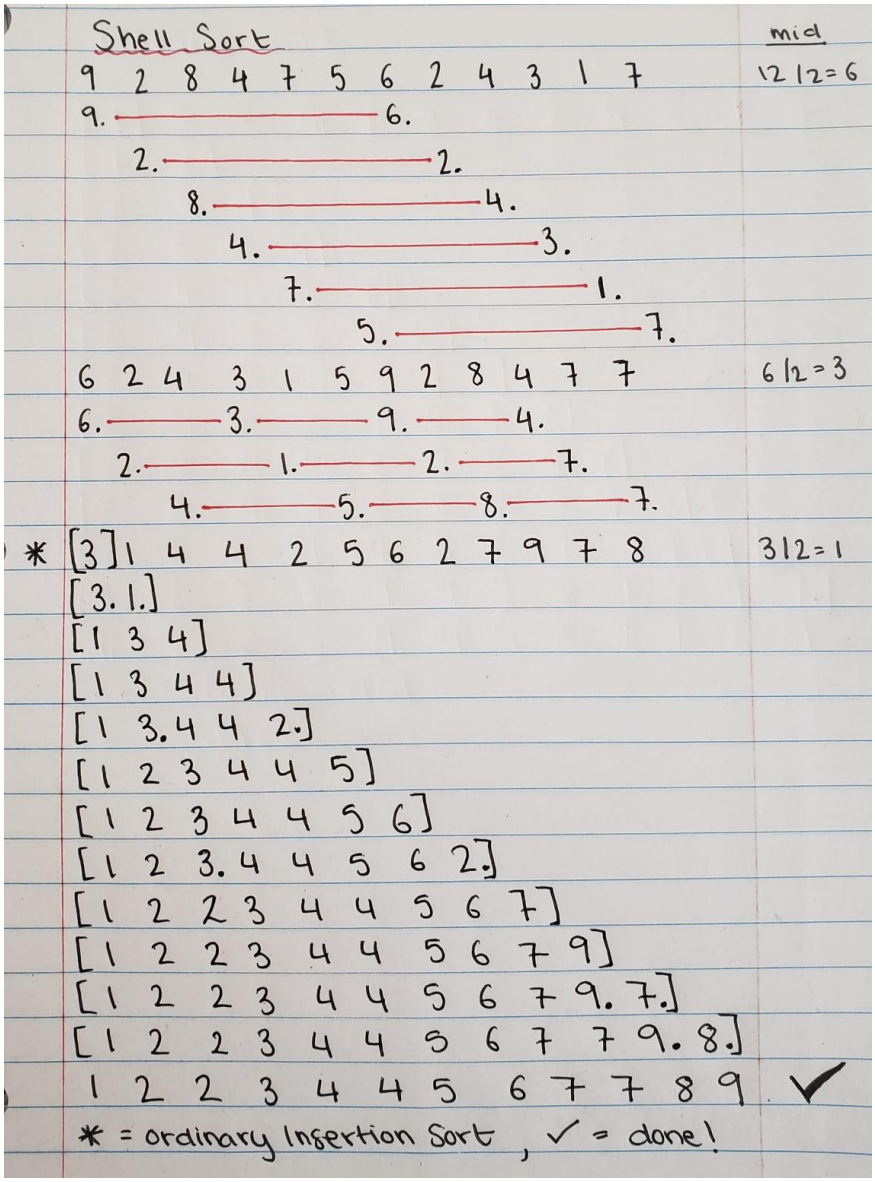
3. Shell Sort

```
ShellSortDriver
/Library/Java/JavaVirtualMachines/
List Before Shell Sort:
9 2 8 4 7 5 6 2 4 3 1 7
Is sorted: false

6 2 8 4 7 5 9 2 4 3 1 7
6 2 8 4 7 5 9 2 4 3 1 7
6 2 4 4 7 5 9 2 8 3 1 7
6 2 4 3 7 5 9 2 8 4 1 7
6 2 4 3 1 5 9 2 8 4 7 7
6 2 4 3 1 5 9 2 8 4 7 7
3 2 4 4 1 5 6 2 8 9 7 7
3 1 4 4 2 5 6 2 8 9 7 7
3 1 4 4 2 5 6 2 7 9 7 8
1 2 2 3 4 4 5 6 7 7 8 9

List After Shell Sort:
1 2 2 3 4 4 5 6 7 7 8 9
Is sorted: true


Process finished with exit code 0
```



Part B:

1. Algorithm A is the time complexity of O(n). It depends on the way and type of sort we use to sort Algorithm B to determine whether Algorithm A or Algorithm B is faster. If we use selection sort to sort Algorithm B, it would take a lot more time and would be less efficient compared to Algorithm A because selection sort is O(n^2) in the both best and worst case for the number of entries in the array. However if we use shell sort in

Algorithm B, it would be faster and more efficient compared to Algorithm A because shell sort is a lot faster compared to selection sort and is O(nlog(n)) in it's best case.

2.

```
ArraySortByFirst
/Library/Java/JavaVirtualMachines/
The array is initially
1  2  3  4  5
3  4  5  1  2
5  2  3  4  1
2  3  1  4  5
4  2  3  1  5

The array after sorting is
1  2  3  4  5
2  3  1  4  5
3  4  5  1  2
4  2  3  1  5
5  2  3  4  1


Process finished with exit code 0
```

Part C:

1.

myQueue.enqueue("Jane");
Queue: Jane
myQueue.enqueue("Jess");
Queue: Jane, Jess
myQueue.enqueue("Jon");
Queue: Jane, Jess, Jon
myQueue.enqueue(myQueue.dequeue());
Queue: Jess, Jon, Jane
myQueue.enqueue(myQueue.getFront());
Queue: Jess, Jon, Jane, Jess
myQueue.enqueue("Jim");
Queue: Jess, Jon, Jane, Jess, Jim

String name = myQueue.dequeue();
Name = Jess
Queue: Jess, Jon, Jane, Jim
myQueue.enqueue(myQueue.getFront());
Queue: Jess, Jon, Jane, Jim, Jess

2.

myDeque.addToFront("Jim");
Queue: Jim
myDeque.addToFront("Jess");
Queue: Jess, Jim
myDeque.addToBack("Jen");
Queue: Jess, Jim, Jen
myDeque.addToBack("Josh");
Queue: Jess, Jim, Jen, Josh
String name = myDeque.removeFront();
Name = Jess
Queue: Jim, Jen, Josh
myDeque.addToBack(name);
Queue: Jim, Jen, Josh, Jess
myDeque.addToBack(myDeque.getFront());
Queue: Jim, Jen, Josh, Jess, Jim
myDeque.addToFront(myDeque.removeBack());
Queue: Jim, Jim, Jen, Josh, Jess
myDeque.addToFront(myDeque.getBack());
Queue: Jess, Jim, Jim, Jen, Josh, Jess

3.

myPriorityQueue.add("Jim");
Queue: Jim
myPriorityQueue.add("Josh");
Queue: Jim, Josh
myPriorityQueue.add("Jon");
Queue: Jim, Josh, Jon
myPriorityQueue.add("Jane");
Queue: Jim, Josh, Jon, Jane
String name = myPriorityQueue.remove();
Name = Jim
Queue: Josh, Jon, Jane
myPriorityQueue.add(name);

Queue: Josh, Jon, Jane, Jim
myPriorityQueue.add(myPriorityQueue.peek());
Queue: Josh, Jon, Jane, Jim, Josh
myPriorityQueue.add("Jose");
Queue: Josh, Jon, Jane, Jim, Josh, Jose
myPriorityQueue.remove();
Queue: Jon, Jane, Jim, Josh, Jose

Part D:

```
DequeTester
/Library/Java/JavaVirtualMachines/jdk-17.0.1.jdk
Empty deque: true
[FRONT] Jerry << Tom << Minnie << Mickey << ...
 [BACK] Sylvester >> Goofy >> Donald >> ...

Empty deque: true
Sayōnara
-> removeFront found deque empty
-> removeBack found deque empty

Process finished with exit code 0
```

Part E:

```
SFSUOneStopClient
/Library/Java/JavaVirtualMachines/jdk-17.0.1.jdk/Contents/Home/
------------------------------------------------------------
             SFSU ONE STOP STUDENT SERVICES CENTER
------------------------------------------------------------
Priority: realistic (provided by supervisors)
          Minnie    Mouse     1001      3.90       10      15
          Daisy     Duck      1003      1.70        1      17
          Donald    Duck      1006      3.10        5       2
          Milo      Dog       1004      3.70        7      17
          Pluto     Dog       1005      3.70        7      17
          Goofy     Dog       1007      2.30       17       1
          Mickey    Mouse     1002      3.70        1      17

Priority: first-name
          Daisy     Duck      1003      1.70        1      17
          Donald    Duck      1006      3.10        5       2
          Goofy     Dog       1007      2.30       17       1
          Mickey    Mouse     1002      3.70        1      17
          Milo      Dog       1004      3.70        7      17
          Minnie    Mouse     1001      3.90       10      15
          Pluto     Dog       1005      3.70        7      17
```

```
Priority: last-name
          Goofy     Dog       1007      2.30       17        1
          Pluto     Dog       1005      3.70        7       17
          Milo      Dog       1004      3.70        7       17
          Daisy     Duck      1003      1.70        1       17
          Donald    Duck      1006      3.10        5        2
          Minnie    Mouse     1001      3.90       10       15
          Mickey    Mouse     1002      3.70        1       17

Priority: student-id
          Minnie    Mouse     1001      3.90       10       15
          Mickey    Mouse     1002      3.70        1       17
          Daisy     Duck      1003      1.70        1       17
          Milo      Dog       1004      3.70        7       17
          Pluto     Dog       1005      3.70        7       17
          Donald    Duck      1006      3.10        5        2
          Goofy     Dog       1007      2.30       17        1

Priority: gpa
          Daisy     Duck      1003      1.70        1       17
          Goofy     Dog       1007      2.30       17        1
          Donald    Duck      1006      3.10        5        2
          Milo      Dog       1004      3.70        7       17
          Pluto     Dog       1005      3.70        7       17
          Mickey    Mouse     1002      3.70        1       17
          Minnie    Mouse     1001      3.90       10       15
```

```
Priority: number-of-small-questions
          Daisy     Duck      1003      1.70        1       17
          Mickey    Mouse     1002      3.70        1       17
          Donald    Duck      1006      3.10        5        2
          Milo      Dog       1004      3.70        7       17
          Pluto     Dog       1005      3.70        7       17
          Minnie    Mouse     1001      3.90       10       15
          Goofy     Dog       1007      2.30       17        1

Priority: number-of-big-questions
          Goofy     Dog       1007      2.30       17        1
          Donald    Duck      1006      3.10        5        2
          Minnie    Mouse     1001      3.90       10       15
          Milo      Dog       1004      3.70        7       17
          Pluto     Dog       1005      3.70        7       17
          Daisy     Duck      1003      1.70        1       17
          Mickey    Mouse     1002      3.70        1       17

Priority: number-of-small-and-big-questions
          Donald    Duck      1006      3.10        5        2
          Goofy     Dog       1007      2.30       17        1
          Daisy     Duck      1003      1.70        1       17
          Mickey    Mouse     1002      3.70        1       17
          Pluto     Dog       1005      3.70        7       17
          Milo      Dog       1004      3.70        7       17
          Minnie    Mouse     1001      3.90       10       15


Process finished with exit code 0
```